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**Preliminary Report**

# **Vlasov–Poisson System for Plasma using the Finite Element Method**

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## Introduction

There are several research projects underway globally, including within the UK, that are focused on simulating plasma in fusion reactors as a potential source of carbon-free energy. Accurately simulating plasma is extremely challenging because the governing partial differential equation — the Vlasov equation — is six-dimensional in phase space, plus time. Specifically, the distribution function depends on both the three spatial coordinates and the three components of particle velocity, resulting in a function  $f(\mathbf{x}, \mathbf{v}, t)$  that must be resolved over a high-dimensional domain.

Due to the computational cost associated with solving such high-dimensional problems, there is considerable focus on the development of “reduced-order modelling” or “model reduction” techniques. These approaches aim to approximate the solution with lower complexity while preserving essential physical behavior, thereby making plasma simulations more tractable and suitable for use in reactor design workflows.

In this project, we explore a hybrid grid–particle method. The spatial part of the distribution function is discretised using a regular grid, which is particularly helpful when simulating complex reactor geometries. Meanwhile, the velocity component is approximated using a particle-based representation. This hybrid scheme effectively maintains an ensemble of distribution functions, where each member describes the density of particles with a specific velocity at every point in space.

To evaluate this approach, we focus on a reduced 1D–1D case (one spatial and one velocity dimension). This lower-dimensional model allows us to investigate the performance and accuracy of the method. The implementation is built using the `firedrakeproject.org` library, which abstracts much of the spatial discretisation, enabling rapid prototyping and experimentation. Subject to performance and results, there may be scope to extend the work to a 2D–2D case using the department’s research computing cluster.

# 1 The Vlasov-Poisson system

The Vlasov–Poisson system describes the evolution of a distribution function  $q(\mathbf{x}, \mathbf{v}, t)$  for a collection of charged particles under their self-consistent electrostatic field. In this “characteristic,” or particle-based, formulation, the continuous distribution is represented as a sum over particle weights and delta functions in velocity space. Each particle weight and velocity then evolves according to transport and acceleration equations, coupled through Poisson’s equation for the electric potential.

## 1.1 Governing Equations

We begin with the general Vlasov–Poisson equation in phase space:

$$\frac{\partial q}{\partial t} + \nabla_{\mathbf{x}} \cdot (\mathbf{v} q) + \nabla_{\mathbf{v}} \cdot (\mathbf{a} q) = 0,$$

where:

- $\nabla_{\mathbf{x}}$  and  $\nabla_{\mathbf{v}}$  denote gradients in physical and velocity space, respectively.
- $\mathbf{v}$  is the particle velocity coordinate driving advection in  $\mathbf{x}$ .
- $\mathbf{a}(\mathbf{x}, t)$  is the acceleration, typically  $-\nabla\phi$  for an electrostatic field.

## 1.2 1D–1V Specialization

In one spatial and one velocity dimension, the equation simplifies to:

$$\frac{\partial q}{\partial t} + \frac{\partial}{\partial x}(v q) + \frac{\partial}{\partial v}(a q) = 0.$$

This form highlights the two coupled advection processes: in  $x$ -space with speed  $v$ , and in  $v$ -space with acceleration  $a$ .

To bridge the continuum formulation and particle methods, we employ an ansatz in which the distribution function is expressed as a sum of Dirac delta–functions located at each particle’s phase–space coordinates.

$$f(\mathbf{x}, \mathbf{v}, t) = \sum_{i=1}^M f_i(\mathbf{x}, t) \delta(\mathbf{v} - \mathbf{v}_i(\mathbf{x}, t)).$$

Here each particle  $i$  carries a weight  $f_i$  and a velocity  $\mathbf{v}_i$ . The delta function in velocity space ensures that all of particle  $i$ ’s contribution is concentrated at its instantaneous velocity.

## 1.3 Poisson coupling

In the Vlasov–Poisson model, the *velocity flux* (or acceleration) is determined directly by the underlying physics of the plasma. For a single-species plasma in which each particle has charge  $q$  and mass  $m$ , the acceleration  $\mathbf{a}$  is given by

$$\mathbf{a} = \frac{q}{m} \mathbf{E}(\mathbf{x}, t), \tag{1}$$

where  $\mathbf{E}(\mathbf{x}, t)$  is the self-consistent electric field. This field is obtained from the electrostatic potential  $\phi(\mathbf{x}, t)$  via

$$\mathbf{E}(\mathbf{x}, t) = -\nabla\phi(\mathbf{x}, t). \tag{2}$$

The potential itself is not externally imposed; rather, it evolves in response to the particle distribution through Poisson’s equation:

$$-\nabla^2\phi(\mathbf{x}, t) = \rho(\mathbf{x}, t), \tag{3}$$

where  $\rho(\mathbf{x}, t)$  denotes the charge density, typically given by  $\rho = qf(\mathbf{x}, t)$  in normalized units. In one spatial dimension, Poisson’s equation simplifies to

$$\frac{d^2\phi}{dx^2} = -\rho(x, t). \quad (4)$$

Hence, the electric field—and thus the velocity flux—is fully determined by the particle distribution function  $f(x, v, t)$ . This self-consistent coupling between microscopic particle motion and the macroscopic field is the essence of **Poisson coupling** in the Vlasov–Poisson system and underlies collective phenomena such as plasma oscillations, wave–particle interactions, and nonlinear instabilities.

## Characteristic Equations for Particles

Substituting the ansatz into the Vlasov equation gives, for each particle  $i$ :

$$\frac{\partial f_i}{\partial t} + \nabla_{\mathbf{x}} \cdot (\mathbf{v}_i f_i) = 0, \quad \frac{\partial \mathbf{v}_i}{\partial t} = \mathbf{a}(\mathbf{x}, t).$$

## 1D Form

In one dimension these read:

$$\frac{\partial f_i}{\partial t} + \frac{\partial}{\partial x}(v_i f_i) = 0, \quad \frac{\partial v_i}{\partial t} = a(x, t).$$

The first is a continuity equation transporting weight  $f_i$  at speed  $v_i$ , and the second evolves the velocity via the force field. Hence the PDE system becomes

$$\partial_t q(x, t) + \partial_x (q(x, t) u(x, t)) = 0, \quad (5)$$

$$- \partial_{xx} \phi(x, t) = q(x, t), \quad (6)$$

$$\partial_t u(x, t) = - \frac{1}{m} \partial_x \phi(x, t), \quad m = 1. \quad (7)$$

## Summary

We have transformed the Vlasov–Poisson PDE into a system of characteristic equations for discrete particle weights and velocities, coupled via Poisson’s equation for the potential. This forms the basis for particle-based and discontinuous-Galerkin solvers in multiple dimensions.

## 2 Finite Element Method: Theoretical Framework

The Finite Element Method (FEM) is a widely used numerical technique for approximating solutions to boundary value problems governed by partial differential equations (PDEs). Its strength lies in its flexibility to handle complex geometries, heterogeneous materials, and general boundary conditions.

### 2.1 Variational Formulation

FEM is rooted in the idea of reformulating a differential equation into a weak (variational) form. For example, consider the model elliptic problem:

$$-\nabla \cdot (\nabla u) = f \quad \text{in } \Omega, \quad u = 0 \quad \text{on } \partial\Omega,$$

where  $\Omega \subset \mathbb{R}^d$  is the computational domain.

Multiplying both sides by a test function  $v \in H_0^1(\Omega)$  and integrating by parts leads to the weak form:

$$\int_{\Omega} \nabla u \cdot \nabla v \, dx = \int_{\Omega} f v \, dx \quad \forall v \in H_0^1(\Omega).$$

This formulation ensures existence and uniqueness under standard assumptions and forms the basis of the finite element discretization.

To obtain a computable approximation, the infinite-dimensional space  $H_0^1(\Omega)$  is replaced by a finite-dimensional subspace  $V_h \subset H_0^1(\Omega)$ , typically composed of *piecewise polynomial functions* defined over a partitioned mesh of the domain.

The discrete variational problem then reads: Find  $u_h \in V_h$  such that

$$\int_{\Omega} \nabla u_h \cdot \nabla v_h \, dx = \int_{\Omega} f v_h \, dx \quad \forall v_h \in V_h.$$

This is the *Galerkin method*, which results in a linear system of equations:

$$A\mathbf{u} = \mathbf{f},$$

where  $A$  is the global stiffness matrix and  $\mathbf{f}$  is the load vector, constructed using the basis of  $V_h$ .

### 2.2 Basis Functions and Mesh Generation

The finite element space  $V_h$  is spanned by a set of *local basis functions*  $\{\phi_i\}$ , usually defined with compact support over individual mesh elements. In the case of linear finite elements (P1), these basis functions are continuous and piecewise linear, taking the value 1 at one mesh node and 0 at all others.

The computational domain is divided into *elements* (e.g., intervals in 1D, triangles or quadrilaterals in 2D), and the quality and resolution of this *mesh* play a crucial role in the accuracy and efficiency of the method.

### 2.3 Numerical Integration and System Assembly

Local contributions to the stiffness matrix and load vector are computed via *numerical quadrature*, often using reference elements. These local matrices are then *assembled* into a global system by leveraging the connectivity of the mesh.

The resulting global system is typically *sparse* and *structured*, which allows for efficient storage and

solution techniques. For further information on finite elements, refer to [1] in this document.

## Finite Element Approximation to Our Problem

We employ an Eulerian description in configuration space coupled with a Lagrangian representation in velocity space, which removes the need for explicit particle-cell searches. In variational form, we seek  $f(x, v, t)$  such that for every smooth test function  $\psi(x, v)$  on the spatial domain  $\Omega \subset \mathbb{R}^n$ ,

$$\frac{d}{dt} \int_{\Omega} \int_{\mathbb{R}^n} \psi f \, dv \, dx - \int_{\Omega} \int_{\mathbb{R}^n} (\nabla_x \psi \cdot v f + \nabla_v \psi \cdot a f) \, dv \, dx = 0.$$

Neglecting collisions, we approximate the solution by a sum of velocity-localized weights,

$$f(x, v, t) = \sum_{i=1}^N f_i(x, t) \delta(v - v_i(x, t)),$$

where each trajectory  $v_i(x, t)$  evolves according to

$$\frac{\partial v_i(x, t)}{\partial t} = a(x, v_i(x, t), t).$$

Substituting this ansatz into the weak form gives

$$\begin{aligned} & \frac{d}{dt} \int_{\Omega} \int_{\mathbb{R}^n} \psi f \, dv \, dx - \int_{\Omega} \int_{\mathbb{R}^n} (\nabla_x \psi \cdot v f + \nabla_v \psi \cdot a f) \, dv \, dx \\ &= \frac{d}{dt} \int_{\Omega} \sum_{i=1}^N \psi(x, v_i(x, t)) f_i(x, t) \, dx \\ & \quad - \int_{\Omega} \sum_{i=1}^N \nabla_v \psi(x, v_i(x, t)) \cdot a(x, v_i(x, t), t) f_i(x, t) \, dx \\ & \quad - \int_{\Omega} \sum_{i=1}^N \nabla_x \psi(x, v_i(x, t)) \cdot v_i(x, t) f_i(x, t) \, dx. \end{aligned}$$

Grouping like terms and noting that the  $\nabla_v \psi$  integrals cancel, we obtain the condition

$$\int_{\Omega} \sum_{i=1}^N \psi(x, v_i(x, t)) \frac{\partial f_i(x, t)}{\partial t} \, dx - \int_{\Omega} \sum_{i=1}^N \nabla_x \psi(x, v_i(x, t)) \cdot v_i(x, t) f_i(x, t) \, dx = 0.$$

This holds precisely when each weight  $f_i$  satisfies its own continuity equation,

$$\frac{\partial f_i}{\partial t} + \nabla_x \cdot (v_i f_i) = 0, \quad i = 1, \dots, N.$$

Finally, the total charge density that enters Poisson's equation is

$$\rho(x, t) = \sum_{i=1}^N f_i(x, t).$$

## References

- [1] Susanne C. Brenner and L. Ridgway Scott. *The Mathematical Theory of Finite Element Methods*. Vol. 15. Texts in Applied Mathematics. Springer, 2008, p. 400.
- [2] Eric Sonnendrücker and Katharina Kormann. *Numerical Methods for Vlasov Equations*. Vol. 107. Lecture Notes. Max-Planck-Institut für Plasmaphysik, 2013, p. 108.